

Answer the following 25 multiple-choice questions. Quiz is 45-minutes (CLOSED-BOOK/NOTE)

Question 1. Which one of the following would not be considered to be a basic operation?

- a. variable initialization
- ☒ b. simple assignment
- c. operations to control loop execution
- d. all of the above

Question 2. When searching an array for a particular value, which case is most useful?

- a. the average case
- ☒ b. the best case
- c. the worst case
- d. none of the above

Question 3. If an algorithm requires 7 basic operations for an algorithm with a problem size of n , the algorithmic complexity is

- ☒ a. $O(1)$
- b. $O(7)$
- c. $O(n)$
- d. $O(7n)$

Question 4. To measure the time requirement of an algorithm, we must

- ☒ a. find an appropriate growth rate function
- b. run the algorithm for different problem sizes
- c. code the algorithm in several different languages and compare the run times
- d. all of the above

Question 5. An algorithm takes 6 seconds to solve a problem of size 100 and 10 minutes to solve a problem of size 1000. What is the likely running time of the algorithm?

- a. Constant
- ☒ b. Linear
- ☒ c. Quadratic
- d. Cubic



Question 6. For large values of n , which statement is true?

- a. $(n^2 + n) / 2$ behaves like n^2
- b. $(n^2 + n) / 2$ behaves like n
- c. $(n^2 + n) / 2$ behaves like $n/2$
- d. none of the above

Question 7. Big O notation describes

- a. the amount of memory required by an algorithm
- b. the difficulty of writing an algorithm to solve a specific problem
- c. an algorithm's efficiency in terms of the work required to solve a problem
- d. the length of an algorithm for solving a specific problem

Question 8. Big O notation is concerned with the growth rate of algorithm run times, so.....

- a. constants are dominant
- b. constants are ignored
- c. constant terms are emphasized
- d. constants with large values are more important than those with low values

Question 9. How many comparisons will the linear search algorithm make if the search key is not in an array of 10 elements?

- a. 0
- b. 10
- c. 9
- d. 5

Question 10: The linear search algorithm runs in time.

- a. quadratic
- b. $O(n)$
- c. constant
- d. nonlinear

Question 11: The worst case in linear search is that every element must be checked to determine whether the search key exists, which occurs if the search key

- a. is the last array element
- b. is the middle element
- c. is the last array element or is not present
- d. None of the above

Question 12: Which of the following statements is true?

- a. The binary search algorithm is less efficient than the linear search, but it requires that the array be sorted
- b. The binary search algorithm is more efficient than the linear search, but it requires that the array be *unsorted*
- ☒ c. The binary search algorithm is more efficient than the linear search, but it requires that the array be sorted
- d. The binary search algorithm is less efficient than the linear search, but it requires that the array be unsorted

Question 13. On average, linear search searches

- a. the whole list
- ☒ b. half of the list
- c. just one element in the list
- d. one-fourth of the list

Question 14. Which of the following statement is false?

- a. Pure quadratic functions contain no linear term ✓
- b. Complete quadratic functions contain a linear term ✓
- ☒ c. If the function is not a pure quadratic function, we can never classify it with the pure quadratic functions
- d. The set of all complexity functions that can be classified with pure quadratic functions is called θn^2

Question 15. Which of the following statement is **FALSE**?

- a. $O(1)$ describes an algorithm that will always execute in the same time (or space) regardless of the size of the input data set. ✓
- b. The Big-O notation of the function $f(n) = (\log n)(n + n^2)$ is $O(n^2 \log n)$. ✓
- c. The Big-O notation measures the best-case complexity of an algorithm. ✓
- ☒ d. Iterative algorithms execute faster than recursion-based ones because no stack needs to be maintained

Question 16: Using a binary search, what is the maximum number of comparisons required to find a search key in a 31-element sorted array?

- ☒ a. 4
- ☒ b. 5
- c. 32
- d. 1

... you have an array of n elements. Suppose you implement Quicksort by always choosing the central element (i.e., not median) of the array as the pivot. In this case, what is the upper bound for the worst case performance?

- a. $O(n \lg n)$
- b. $O(n^3)$
- ☒ c. $O(n^2)$
- d. $O(\lg n)$

Q2. Which of the following method is the most effective for picking the pivot element for Quicksort?

- a. The element in the first position
- b. The element with the smallest value
- c. A randomly selected element
- ☒ d. The median of the elements

Q3. What is the average running time of a Quicksort algorithm?

- a. $O(N^2)$
- b. $O(N)$
- ☒ c. $O(N \lg N)$
- ☒ d. $O(\lg N)$

Q4. You must merge two sorted lists of size a and b , respectively. What is the number of comparisons needed in the worst case by the Merge algorithm?

- a. $a * b$
- b. $a + b$
- c. $\max(a, b)$
- ☒ d. $a + b - 1$

Q5. What is the worst-case time complexity of the Mergesort algorithm to sort an array of n records?

- ☒ a. $\Theta(n^2)$
- b. $\Theta(n)$
- c. $\Theta(n \lg n)$
- d. $\Theta(\lg n)$

Q6. When selecting a pivot value in Quick sort algorithm, a simple thing to do is to always pick from the same place in the partition. If we use this approach, does it matter whether we always pick from the first position in the partition, the last position in the partition, or the middle position in the partition?

- ☒ a. It is much better to pick the first value
- b. It is much better to pick the last value
- c. It doesn't really matter, they are all equally good or bad
- d. It is much better to pick the middle value

Q7. Which of the following statement is false?

- a. Divide-and-conquer algorithms should be avoided if an instance of size n is divided into two or more instances each almost of size n
- b. In the multiplication of large integers the division process is continued until a threshold value is reached
- ☒ c. In quick sort, all items smaller than the pivot are placed on the right of it
- d. Divide-and-conquer algorithms should be avoided if an instance of size n is divided into almost n instances of size n/c , where c is a constant

Q8. Which of the following statement is false?

- a. We can represent a large integer by using an array of integers, in which each array slot stores one digit
- ☒ b. Strassen's matrix multiplication method requires more multiplications but fewer additions/subtractions than the standard matrix multiplication method
- c. In Binary Search, the solution of the original instance is the solution to the smaller instance.
- d. The divide-and-conquer approach is a top-down approach

Q9. Let P be a Quick Sort program to sort numbers in ascending order using the first element as pivot. Let t_1 and t_2 be the number of comparisons made by P for the inputs $\{1,2,3,4,5\}$ and $\{4,1,5,3,2\}$ respectively. Which of the following holds true?

- a. $t_1 = 5$
- b. $t_1 < t_2$
- ☒ c. $t_1 > t_2$
- d. $t_1 = t_2$

Q10. Suppose we have a $O(n)$ algorithm that finds median of an unsorted array. Now consider a QuickSort implementation where we first find median using the above algorithm, then use median as pivot. What is the worst case time complexity of this modified QuickSort?

- a. $O(n^2 \lg n)$
- ☒ b. $O(n^2)$
- c. $O(n \lg n \lg n)$
- ☒ d. $O(n \lg n)$

Answer the following 20 multiple-choice questions. Quiz is 45 minutes.

Question 1. Which of the following gives the worst case complexity of the Quicksort? (i.e., the answer must be given using the complexity of the time to partition, time to sort left subarray, and the time to sort right subarray)

- a. $W(n) = (n) + W(0) + W(n)$
- b. $W(n) = (n - 2) + W(1) + W(n)$
- ☒ c. $W(n) = (n - 1) + W(0) + W(n-1)$
- d. $W(n) = (n - 1) + W(1) + W(n-1)$

Question 2. What is the every case time complexity for the binary search algorithm?

- a. n
- b. n^2
- c. $n/2$
- ☒ d. There is no every case time complexity

Question 3. Assume that you have an array of n elements. Suppose you implement Quicksort by always choosing the central element (i.e., not median) of the array as the pivot. In this case, what is the upper bound for the worst case performance?

- a. $O(n \log n)$
- b. $O(n^3)$
- ☒ c. $O(n^2)$
- d. $O(\log n)$

Question 4. Which of the following is the every case time complexity of Strassen algorithm in terms of number of multiplications?

- | | |
|---|-------------------------|
| a. $T(n) = 7 T(n/2) + 18(n/2)^2$ | $n > 1, n$ a power of 2 |
| ☒ b. $T(n) = 7 T(n/2)$ | $n > 1, n$ a power of 2 |
| c. $T(n) = 7 T(n) + 18n$ | $n > 1, n$ a power of 2 |
| d. $T(n) = 7 T(n) + 18$ | $n > 1, n$ a power of 2 |

Question 5. In the Strassen algorithm itself, how many times is the Strassen algorithm called recursively?

- a. 1
- b. 3
- ☒ c. 5
- d. 7

Question 12. When selecting a pivot value in Quick sort algorithm, a simple thing to do is to always pick from the same place in the partition. If we use this approach, does it matter whether we always pick from the first position in the partition, the last position in the partition, or the middle position in the partition?

- a. It is much better to pick the first value
- b. It is much better to pick the last value
- c. It doesn't really matter, they are all equally good or bad
- d. It is much better to pick the middle value

Question 13. Which of the following statement is false?

- a. Divide-and-conquer algorithms should be avoided if an instance of size n is divided into two or more instances each almost of size n
- b. In the multiplication of large integers the division process is continued until a threshold value is reached
- c. In quick sort, all items smaller than the pivot are placed on the right of it
- d. Divide-and-conquer algorithms should be avoided if an instance of size n is divided into almost n instances of size n/c , where c is a constant

Question 14. How can you apply the Strassen matrix multiplication algorithm if n is not a power of 2?

- a. Remove one column to the original matrices to make the dimension a power of 2
- b. Remove one row to the original matrices to make the dimension a power of 2
- c. Add sufficient numbers of columns and rows of 0s to the original matrices to make the dimension a power of 2
- d. None of the above

Question 15. Which of the following statement is false?

- a. We can represent a large integer by using an array of integers, in which each array slot stores one digit
- b. Strassen's matrix multiplication method requires more multiplications but fewer additions/subtractions than the standard matrix multiplication method
- c. In Binary Search, the solution of the original instance is the solution to the smaller instance.
- d. The divide-and-conquer approach is a top-down approach

Question 16. Let P be a Quick Sort program to sort numbers in ascending order using the first element as pivot. Let t_1 and t_2 be the number of comparisons made by P for the inputs $\{1,2,3,4,5\}$ and $\{4,1,5,3,2\}$ respectively. Which of the following holds true?

- a. $t_1 = 5$
- b. $t_1 < t_2$
- c. $t_1 > t_2$
- d. $t_1 = t_2$

Question 17. Suppose we have a $O(n)$ algorithm that finds median of an unsorted array. Now consider a QuickSort implementation where we first find median using the above algorithm, then use median as pivot. What is the worst case time complexity of this modified QuickSort?

- a. $O(n^2 \lg n)$
- ☒ b. $O(n^2)$
- c. $O(n \lg n \lg n)$
- d. $O(n \lg n)$

Question 18. Which of the following statement is FALSE?

- a. Iterative algorithms execute faster than recursion because no stack needs to be maintained ✓
- b. A sorting algorithm that does not use any extra space beyond that needed to store input is called an in-place sorting algorithm ✓
- c. In Strassen's method, large matrices are usually divided into four sub-matrices ✓
- ☒ d. Merge sort is a sorting technique that sequences a list by continually dividing the list and then moving the lower items to one side and the higher items to the other side

Question 19. Which of the following statements is FALSE?

- a. The average running time of a Quick Sort is $O(N \log N)$
- b. The worst case complexity of Merge Sort is $O(N \log N)$
- ☒ c. In Quick Sort, it is a good choice to select the first item as the pivot item if the array is already sorted
- d. Strassen's matrix multiplication requires fewer multiplications than the standard matrix multiplication method

Question 20. Which of the following is the every case time complexity of Strassen algorithm in terms of number of additions/subtractions?

- a. $T(n) = 7 T(n/2) + 18(n/2)^2$
- b. $T(n) = 7 T(n/2)$
- c. $T(n) = 7 T(n) + 18n$
- ☒ d. $T(n) = 7 T(n) + 18$

- $n > 1, n$ a power of 2
- $n > 1, n$ a power of 2
- $n > 1, n$ a power of 2
- $n > 1, n$ a power of 2

Question 6. Which of the following methods is the most effective for picking the pivot element for Quicksort algorithm?

- a. The element in the first position
- b. The element with the smallest value
- c. A randomly selected element
- ☒ d. The median of the elements

Question 7. What is the average running time of a Quicksort algorithm?

- a. $O(N^2)$
- b. $O(N)$
- ☒ c. $O(N \log N)$
- d. $O(\log N)$

Question 8. You must merge two sorted lists of size a and b, respectively. What is the number of comparisons needed in the worst case by the Merge algorithm?

- a. $a * b$
- b. $a + b$
- c. $\max(a, b)$
- ☒ d. $a + b - 1$

Question 9. If the Divide and Conquer approach is applied for naïve (traditional) Matrix Multiplication (i.e., treating the input $n \times n$ matrices as block 2×2 matrices), how many times is the Matrix Multiplication algorithm called recursively?

- a. 1
- b. 2
- ☒ c. 4
- d. 8

Question 10. What is the worst-case time complexity of the Merge sort algorithm to sort an array of n records?

- ☒ a. $\Theta(n^2)$
- b. $\Theta(n)$
- c. $\Theta(n \lg n)$
- d. $\Theta(\lg n)$

Question 11. Which of the following is the every case time complexity of the Partition algorithm used in Quicksort?

- a. n
- ☒ b. $n - 1$
- c. $n - 2$
- d. n^2